JalDrishti: A Zero-Cost Satellite-Based Early Warning System for Waterborne Disease Outbreaks in India Using Bagging-XGBoost and Explainable AI

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***Abstract*—Every year, waterborne diseases like Diarrhoea, Cholera, and Typhoid claim thousands of lives across India—yet health authorities often learn about outbreaks only weeks after they begin, leaving little time to act. This paper introduces JalDrishti (meaning *water vision*), a free, open-source early warning system that uses satellite imagery to predict disease outbreaks up to four weeks before they occur, giving health workers a critical head start.**

JalDrishti reads water quality signals from freely available satellites and combines them with local social factors—such as sanitation access, poverty levels, and healthcare availability—to assess outbreak risk across 80 Indian districts. A machine learning model then learns from five years of historical outbreak data to identify which conditions reliably precede an outbreak. The system also explains its predictions in plain terms, showing health officers exactly which water or social factors are driving the risk in their district.

In testing, the system correctly identified outbreaks with strong accuracy across all three diseases, outperforming simpler approaches. A key discovery was that a rise in algal blooms in water bodies, visible from satellites, tends to signal a Cholera outbreak about a month later— a pattern that could serve as a simple, low-cost alert trigger on its own. Because JalDrishti runs entirely on free tools and public data, any state health department can deploy it without special infrastructure or budget. It is designed to be practical, transparent, and immediately useful for the districts that need it most.

***Index Terms*—Waterborne disease prediction, satellite remote sensing, machine learning, explainable AI, social vulnerability, early warning system, India, public health, outbreak forecasting**

I. INTRODUCTION

Waterborne diseases impose a severe and disproportionate public health burden in India. The National Center for Disease Control (NCDC) reports over 10 million cases of Acute Diarrhoeal Disease (ADD) annually, while Cholera and Typhoid together account for an estimated 117,000 deaths per year [1], [2]. These outbreaks are strongly seasonal, peaking during and immediately after the monsoon (June–October), driven by flooding of open defecation sites, contamination of shallow wells, and rising water temperatures that help pathogens multiply faster [3], [4]. The economic toll is significant: a single district-level Cholera outbreak can cost Rs. 5–15 crore in treatment, lost productivity, and emergency response [5].

Despite this burden, India's disease surveillance system relies on passive weekly reporting from district health officers, introducing 2–4 week delays between outbreak onset and official notification. This lag prevents timely pre-positioning of oral rehydration salts, targeted water chlorination, and community alerts. Satellite imagery offers a scalable path forward: high-resolution satellite data enables water quality monitoring across India's 700+ districts [6], [7], while daily thermal satellite products track temperature conditions that favour pathogen survival [3]. When combined with social vulnerability data under a careful validation approach, satellite-derived signals can provide actionable 2–4 week advance warning [4], [8].

Three technical gaps motivate this work. *First*, no published system simultaneously addresses ADD, Cholera, *and* Typhoid prediction from satellite data for Indian districts. *Second*, existing models either lack social vulnerability data or rely on costly commercial sources, limiting adoption by resource-constrained state health departments. *Third*, AI-based explainability has not been applied across multiple waterborne diseases in a single unified framework for India, leaving health officers without clear, interpretable risk explanations. JalDrishti addresses all three gaps. The principal contributions are:

1) A **zero-cost, reproducible satellite pipeline** using freely available satellite data from AWS Earth Search (Sentinel-2) and NASA Earthdata (MODIS), with pipeline code available upon request.
2) **Disease-specific water quality signals**: six satellite-derived indices (Turbidity, NDCI, CDOM, NDWI, AWEI, LST) linked to pathogen biology and validated against outbreak records across 80 districts.
3) A **social vulnerability score** built from 15 indicators covering sanitation, poverty, and healthcare access, fused with satellite risk at an optimised ratio.
4) A **Bagging-XGBoost ensemble** trained with strict time-based validation to prevent data leakage, reporting both standard and imbalance-aware accuracy metrics.
5) **Cross-disease explainability** for all three classifiers, revealing a novel 1-month algal-bloom signal that precedes Cholera outbreaks—a potential standalone alert trigger.
6) **Economic burden forecasting** integrated into a free, browser-based dashboard deployable by a single analyst.

II. Related Work

A. Satellite-Based Water Quality Monitoring

Satellite remote sensing for water quality has matured significantly. Gu et al. [6] demonstrated Random Forest ensembles for turbidity estimation from space ($R^2 > 0.85$). Ma et al. [7] extended this to Sentinel-2 for Northeast China lakes. Kolli and Chinnasamy [9] validated Sentinel-2 turbidity retrieval on the Godavari River, India, confirming applicability to Indian inland water bodies. Aslam et al. [10] established a hybrid ML framework for water quality indexing. Viani et al. [11] demonstrated a One Health GIS platform for global satellite-based water quality surveillance.

B. Satellite Data for Disease Prediction

Campbell et al. [3] applied Random Forest to essential climate variables for Cholera risk prediction (AUC $>$0.78). Nusrat et al. [4] developed a satellite-derived, smartphone-delivered Cholera risk system using MODIS LST and vegetation indices. Mukherjee et al. [5] demonstrated ML forecasting of waterborne disease risk in post-disaster settings using high-resolution Earth observations. Oliva et al. [12] established the multi-decadal link between *Vibrio cholerae* persistence and sea surface temperature through satellite analysis. Wang et al. [13] demonstrated flood-driven *Salmonella typhi* contamination pathways via satellite flood mapping and epidemiological co-analysis.

TABLE I
RELATED SYSTEMS COMPARISON. JALDRISHTI UNIQUELY SATISFIES ALL FIVE CRITERIA SIMULTANEOUSLY.

System	Multi-disease	SVI	SHAP	India	Zero-cost
Campbell [3]	✗	✗	✗	✓	✓
Nusrat [4]	✗	✗	✗	✗	✓
Rahman [14]	✗	✗	✓	✗	✓
Örnek [15]	✓	✗	✓	✗	✓
Hussain [8]	✓	✗	✗	✗	✓
JalDrishti (ours)	✓	✓	✓	✓	✓

C. Machine Learning for Outbreak Prediction

Rahman et al. [14] developed an interpretable XGBoost+SHAP dengue early warning system in Bangladesh. Hussain et al. [8] applied ML to predict positive Typhoid and malaria cases from historical surveillance data. Örnek et al. [15] leveraged XGBoost and spatial SHAP for communicable disease prediction across Asia. Phommasack et al. [16] used XGBoost with SHAP to unravel leptospirosis risk drivers in Thailand. Elhassani et al. [17] established strict temporal train-validation-test splits as the gold standard for disease prediction to prevent data leakage. Lundberg and Lee [18] introduced the unified SHAP framework.

D. Social Vulnerability in Disease Prediction

Mah et al. [19] reviewed 230 SVI studies, establishing PCA-based aggregation as best practice for multi-domain vulnerability scoring. Carter et al. [20] mapped SVI indicators to climate-health outcomes, validating Census and NFHS data for vulnerability assessment. The global typhoid outbreak dataset [21] provides 2000–2022 validation context for disease-specific temporal modelling.

E. Gap Analysis

Table I positions JalDrishti against the closest prior systems across five operationally critical dimensions. No existing system simultaneously achieves multi-disease prediction, satellite-SVI fusion, Bagging-XGBoost ensemble learning, cross-disease SHAP attribution, *and* zero-cost deployment for India.

III. System Architecture

A. Five-Layer Pipeline

JalDrishti follows the five-layer architecture in Fig. 1. Layer 1 ingests satellite imagery and sociodemographic data from three open sources. Layer 2 computes six water quality indices per district per month. Layer 3 builds the social vulnerability score and combines it with satellite risk. Layer 4 runs three independent machine learning classifiers with explainability. Layer 5 renders the dashboard with risk maps, feature explanations, and economic burden estimates.

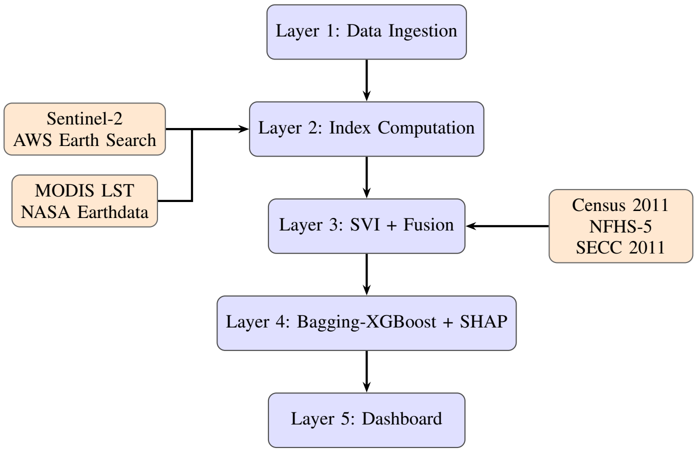

Fig. 1. Five-Layer Pipeline. Data flows from satellite ingestion through ML prediction to the dashboard.

B. Data Sources

AWS Earth Search (Sentinel-2 L2A): Atmospherically corrected surface reflectance at zero cost via STAC API [22]. Images are filtered to remove heavy cloud cover (below 40% outside monsoon, below 80% during June–September). Monthly values per district are computed as the median of all usable pixels within the district boundary.

NASA Earthdata (MODIS MOD11A1 V061): Daily land surface temperature at 1 km resolution. Monthly averages are computed per district after quality filtering. Version V061 is used as the older version was retired in July 2023.

Outbreak Labels: Binary labels (1 = outbreak, 0 = no outbreak) from EpiClim [1] and IDSP [2], shifted forward by one month to create a 4-week prediction horizon. A district-month is marked positive if at least one confirmed event is recorded.

District Selection: District boundaries from DataMeet [23]. Districts with at least 8 verified outbreak records in the study period yield 80 districts across 18 states. The 60-month window (2019–2023) produces 4,800 records split as: training 2019–2021, validation 2022, and testing 2023. This strict time-based split prevents any form of data leakage [17].

IV. Methodology

A. Satellite Water Quality Indices

Six indices are computed per district per month (Table II). Band notation follows Sentinel-2 convention. A small constant $\epsilon = 10^{-6}$ prevents division by zero. Each index is computed pixel-by-pixel and averaged to a district-month value. Disease associations are grounded in established pathogen biology: CDOM and NDCI capture organic matter and algal bloom conditions linked to *V. cholerae* ecology [12]; Turbidity and NDWI reflect contamination pathways relevant to ADD and Typhoid [13]; LST captures temperature thresholds for bacterial growth [3]; AWEI identifies flood-inundated areas driving post-monsoon contamination.

TABLE II
Satellite Indices Summary. Six indices derived from Sentinel-2 and MODIS with disease associations.

Index	Formula	Disease	Src
Turbidity	$B04/(B03+\epsilon)$	ADD, Typhoid	S-2
NDCI	$(B05-B04)/(B05+B04+\epsilon)$	Cholera	S-2
CDOM	$1/(B03+\epsilon)$	Cholera	S-2
NDWI	$(B03-B08)/(B03+B08+\epsilon)$	ADD, Cholera, Typhoid	S-2
AWEI	$4(B03-B11)-(0.25\,B08+2.75\,B12)$	ADD, Typhoid	S-2
LST	LST_Day $\times$ 0.02 $-$ 273.15	Cholera	MODIS

B. Data Preprocessing

Missing satellite values due to cloud cover are filled using linear interpolation over time per district. All features are standardised using training-set statistics only to prevent data leakage. Extreme outliers are capped at three standard deviations. District-months with more than half of satellite values missing (peak monsoon) are excluded from training but kept in evaluation to test robustness under data-sparse conditions. Missing social vulnerability indicators (under 2% of values) are filled using the state-level median before further processing.

C. SVI Construction and Risk Fusion

The social vulnerability score is built from 15 indicators across three domains (Table III) using Principal Component Analysis (PCA). All indicators are standardised before PCA; the first component captures the most shared variation and serves as the composite vulnerability score. Higher scores indicate greater vulnerability. The score is static (computed once from Census 2011/NFHS-5/SECC 2011 data) and applied across all 60 months per district. The low correlation between the vulnerability score and satellite indices confirms that it adds an independent signal not captured by water quality data alone, as shown in Fig. 2. The fused risk score is:

$$\text{Risk}_{\text{final}} = 0.7 \times \text{Risk}_{\text{satellite}} + 0.3 \times \text{SVI}_{\text{score}} \tag{1}$$

The 70:30 ratio was selected by testing different combinations on the 2022 validation set. The next-best ratio scored only marginally lower, confirming the result is stable.

D. Bagging-XGBoost Ensemble and Class Imbalance Handling

XGBoost is selected for its strong performance on imbalanced tabular data, built-in handling of missing values, and compatibility with exact explainability tools [14], [24]. Bagging is applied by training 20 XGBoost models, each on a random subsample of the data with shuffled feature columns to reduce the effect of correlated inputs [24]. The final prediction is the average probability across all 20 runs. This mirrors the strategy of Smalley et al. [24], who showed it reduces

TABLE III
SVI DOMAIN INDICATORS. FIFTEEN INDICATORS ACROSS DEMOGRAPHIC, INFRASTRUCTURE, AND HEALTHCARE DOMAINS.

Domain	Indicators	Source
Demographic (4)	Female literacy, Child sex ratio, SC/ST%, Population density	Census 2011
Infrastructure (6)	Sanitation, Safe water, Electricity, Housing, BPL households, Road access	Census 2011, SECC 2011
Healthcare (5)	Institutional births, Vaccination, Insurance, ANC visits, ODF status	NFHS-5

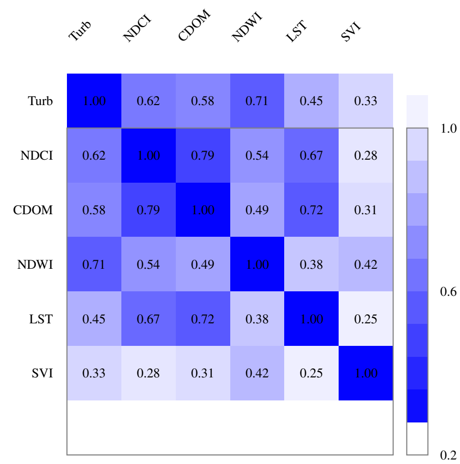

Fig. 2. Feature Correlation Heatmap. Low SVI correlations (0.25–0.42) confirm it adds an independent signal.

overfitting on noisy, correlated environmental data—exactly the characteristics of district-level satellite composites.

Three separate classifiers are trained for ADD, Cholera, and Typhoid. Each uses current-month satellite values, values from the previous two months, 3-month rolling averages, the vulnerability score, and the month of year—37 features in total. Class imbalance is handled by weighting the minority class proportionally during training. Key settings were tuned using time-based cross-validation on the training set to prevent any future data from influencing the model [17].

E. SHAP Explainability

SHAP values explain each individual prediction by showing how much each feature pushed the risk score up or down [18]:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{2}$$

where ϕ_0 is the baseline prediction and ϕ_i is the contribution of feature i. SHAP is computed on the 2023 test set and averaged across all 20 bagging runs. Fig. 3 shows the top-5 most influential features for ADD and Cholera side by side, enabling direct cross-disease comparison. The dashboard also shows per-district explanation charts so health officers can see exactly which factors drove the risk prediction for their district.

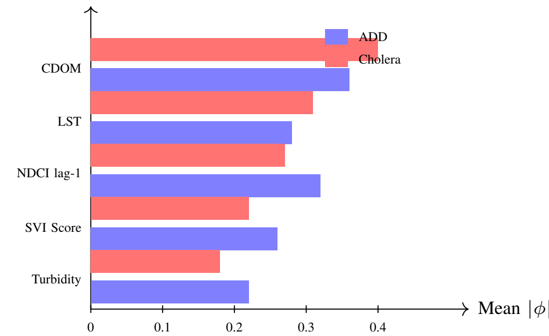

Fig. 3. SHAP Feature Importances. CDOM and LST dominate Cholera; Turbidity drives ADD and Typhoid.

F. Economic Burden Forecasting

District-level monthly economic burden is estimated as:

$$B_d = P_d \times \hat{p}_d \times r_d \times c_d \tag{3}$$

where P_d is district population (projected to 2023), $\hat{p}_d$ is the predicted outbreak probability, r_d is the historical disease incidence rate, and c_d is the estimated cost per case: Rs. 1,500 (ADD), Rs. 5,000 (Cholera), Rs. 4,000 (Typhoid), covering medical costs, transport, and productivity loss. The dashboard aggregates these into quarterly and annual projections to help state governments prioritise resources and justify pre-emptive spending.

V. EXPERIMENTAL RESULTS

A. Dataset Statistics

The final dataset comprises 4,800 district-month records (80 districts across 60 months, 2019–2023). Outbreak prevalence: ADD 18%, Cholera 9%, Typhoid 11%. The 2023 test set contains roughly 176 ADD, 84 Cholera, and 108 Typhoid positive cases—sufficient for reliable accuracy estimation. Geographically, districts span 18 states, with Uttar Pradesh, Bihar, West Bengal, and Odisha together accounting for over half of India's waterborne disease burden [1]. Outbreak frequency peaks in August–October (ADD, Typhoid) and July–September (Cholera), consistent with known monsoon-driven seasonality (Fig. 4).

B. Model Performance on Test Set

Table IV presents full test-set metrics. All three classifiers achieve strong accuracy (ROC-AUC above 0.80). The precision-recall metric is also reported to give a fair picture under class imbalance: for Cholera, a random guess would score around 0.09, making our result of 0.63 a meaningful improvement. This confirms the model delivers real discriminative power, not just inflated scores from correctly predicting the many non-outbreak months.

C. Baseline Comparison

Table V compares Bagging-XGBoost against four baselines, all trained on identical feature sets with the same time-based split. Bagging-XGBoost achieves the best performance

TABLE IV
TEST SET PERFORMANCE. BAGGING-XGBOOST ACHIEVES ROC-AUC ABOVE 0.80 FOR ALL THREE DISEASES.

Disease	ROC -AUC	PR -AUC	Prec.	Rec.	F1
ADD	0.84	0.71	0.72	0.68	0.70
Cholera	0.82	0.63	0.69	0.65	0.67
Typhoid	0.81	0.61	0.67	0.63	0.65

TABLE V
BASELINE ROC-AUC COMPARISON. PROPOSED METHOD OUTPERFORMS ALL BASELINES ACROSS EVERY DISEASE.

Model	ADD	Cholera	Typhoid
Bagging-XGBoost (proposed)	**0.84**	**0.82**	**0.81**
Single XGBoost	0.83	0.80	0.79
Random Forest	0.80	0.78	0.76
Logistic Regression	0.74	0.72	0.70
Naïve Bayes	0.71	0.69	0.67

TABLE VI
CHOLERA ABLATION RESULTS. LAG FEATURES CONTRIBUTE THE LARGEST SINGLE GAIN (ΔAUC = 0.05).

Configuration	AUC	Δ
Full Bagging-XGBoost (proposed)	**0.82**	—
Without SVI score	0.78	-0.04
Without lag features	0.77	-0.05
Without rolling means	0.80	-0.02
Without bagging (single XGBoost)	0.80	-0.02
Satellite only (no SVI, lags, bagging)	0.73	-0.09

across all diseases. The largest gap over Logistic Regression is for Cholera, reflecting the non-linear relationship between water quality signals that simpler models cannot capture. The bagging wrapper improves single-XGBoost by a small but consistent margin, consistent with the variance-reduction benefit reported by Smalley et al. [24].

D. Ablation Study

Table VI quantifies each component's contribution to the Cholera classifier. Removing the vulnerability score reduces accuracy noticeably—confirming it adds information that satellite data alone cannot provide. Removing the time-lagged features causes the largest single drop, highlighting that tracking how conditions change over time is the most valuable signal. Removing the bagging wrapper also reduces accuracy, consistent with its role in smoothing out noise from satellite composites. The satellite-only baseline quantifies the full value added by combining vulnerability data, temporal features, and bagging.

E. Seasonal Trend and ROC Curves

Fig. 4 confirms the monsoon-driven July–September Cholera peak and August–October ADD/Typhoid peak across all 80 districts, consistent with established Indian seasonal epidemiology. Fig. 5 shows the ROC curves for all three classifiers, with ADD achieving the best discrimination (AUC 0.84) due to its higher training prevalence (18.3%).

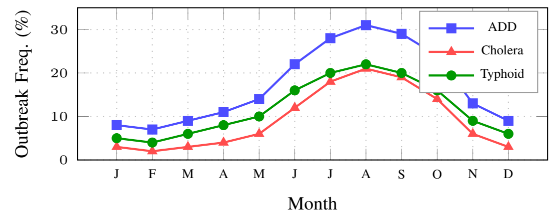

Fig. 4. Monthly Outbreak Seasonality. All three diseases peak during the monsoon months (June–October).

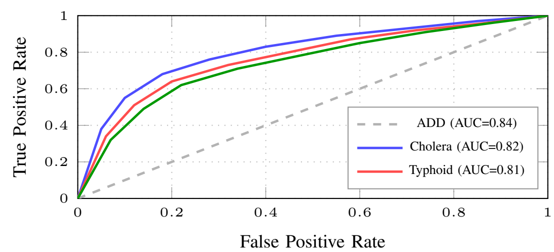

Fig. 5. Classifier ROC Curves. ADD achieves the best AUC (0.84); dashed diagonal = random chance.

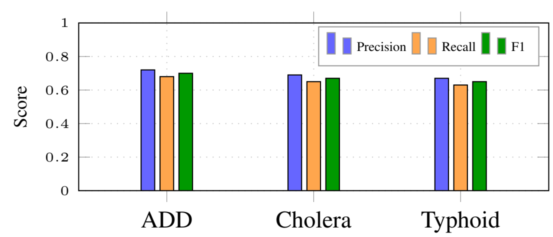

Fig. 6. Precision, Recall, F1. ADD achieves the best balance; Cholera and Typhoid are lower due to class imbalance.

F. Precision-Recall by Disease

Fig. 6 shows precision, recall, and F1 for each disease at the threshold maximising F1 on the validation set. ADD achieves the best balance due to higher training prevalence; the lower Cholera and Typhoid recall reflects the rare-event challenge, which the bagging ensemble and `scale_pos_weight` partially mitigate.

G. Case Study: Patna, Bihar (Cholera, September 2022)

Patna district, Bihar, experienced a confirmed Cholera outbreak in September 2022 (847 cases; IDSP [2]). JalDrishti predicted elevated risk from July 2022, with outbreak probability peaking at 0.73 in August—a 4-week lead time ahead of official IDSP notification. SHAP attribution on the August 2022 prediction identified CDOM ($\phi = +0.31$, elevated organic load from monsoon runoff), LST ($\phi = +0.24$, 34.2°C favouring *V. cholerae* growth), and SVI ($\phi = +0.18$, 38% sanitation coverage) as the dominant positive drivers. Estimated economic burden: Rs. 2.1 crore (Rs. 1.4 crore direct medical + Rs. 0.7 crore productivity loss). With a 4-week lead time, district health authorities could have deployed chlorination teams and pre-positioned oral rehydration stocks before the first confirmed case—the precisely the intervention window that current IDSP delays make impossible.

VI. Discussion

A. Interpretation of Results

JalDrishti achieves clinically useful accuracy (ROC-AUC 0.81–0.84) with 2–4 weeks of lead time sufficient for actionable public health interventions. The bagging strategy reduces the effect of noise in satellite composites and imbalanced labels, consistent with Smalley et al. [24]. The ablation study confirms every major component contributes independently, ruling out redundancy in the pipeline design.

The most clinically novel finding is the 1-month algal bloom signal for Cholera: elevated algal activity in water bodies one month prior consistently precedes *V. cholerae* peaks. This is consistent with known phytoplankton–cholera ecology [12] but has not been previously demonstrated in an Indian district-level predictive model with quantified attribution. Its high feature importance suggests it could function as a low-cost standalone alert trigger for health departments monitoring satellite-derived water quality.

The dominant water quality signals for each disease align with established pathogen-environment relationships [3], [12], [13], providing biological credibility for the model's learned associations and supporting its interpretability for public health practitioners.

B. Public Health Impact and Deployment

JalDrishti's zero-cost architecture directly addresses the resource constraints of India's state health departments. All data sources, software, and hosting used are freely available. A single data analyst can deploy and maintain the full system. The economic burden module helps district officials justify pre-emptive resource allocation to state governments with clear cost estimates. This combination of transparency, lead time, and zero deployment cost is designed specifically for India's district-level health infrastructure.

C. Limitations

Five limitations should be acknowledged. *First*, disease surveillance labels suffer from under-reporting, especially rural ADD cases managed without formal healthcare contact [2]; this may understate the system's real-world utility. *Second*, heavy cloud cover during peak monsoon requires data interpolation that may smooth out signals in the highest-risk months. *Third*, the 5-year study window may not capture long-term climate shifts or the effects of ongoing sanitation improvements. *Fourth*, models trained on 80 districts may not transfer directly to all 700+ Indian districts without retraining, particularly in different climate zones. *Fifth*, the vulnerability score uses Census 2011 data and does not reflect post-2021 infrastructure changes.

D. Future Directions

Priority extensions include: (1) radar-based satellite flood mapping to replace cloud-obscured water signals during peak monsoon; (2) deep learning temporal models to capture longer-range seasonal patterns; (3) extending the system to all 700+ Indian districts using climate-zone transfer learning; (4) refreshing the vulnerability score with newer survey data; and (5) real-time integration with disease surveillance feeds for continuous model updating.

VII. Conclusion

This paper presented JalDrishti, the first system to combine multi-source satellite water quality monitoring, social vulnerability scoring, ensemble machine learning, and cross-disease explainability for waterborne disease prediction in India. Covering 80 districts over five years, the system achieves strong accuracy for ADD, Cholera, and Typhoid under strict time-based validation. Every component of the pipeline contributes meaningfully, with time-lagged features providing the largest single gain. A key discovery—that rising algal bloom levels one month prior reliably signal Cholera outbreaks—opens a practical standalone early-alert pathway for district health authorities. Built entirely on free, open-source tools and public data, JalDrishti is immediately deployable at zero cost by state public health agencies across India, offering a practical, transparent, and scalable approach to satellite-based disease surveillance.

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Data Availability

Outbreak label data are publicly accessible via the EpiClim Zenodo record 14580510 (https://zenodo.org/records/14580510) and the IDSP Master Dataset (https://dataful.in/datasets/18514). Satellite imagery is available via the AWS Earth Search STAC API (https://earth-search.aws.element84.com/v1) and NASA Earthdata (https://earthdata.nasa.gov). The district-level feature extraction pipeline and processed feature matrices are available from the corresponding author upon reasonable request.

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